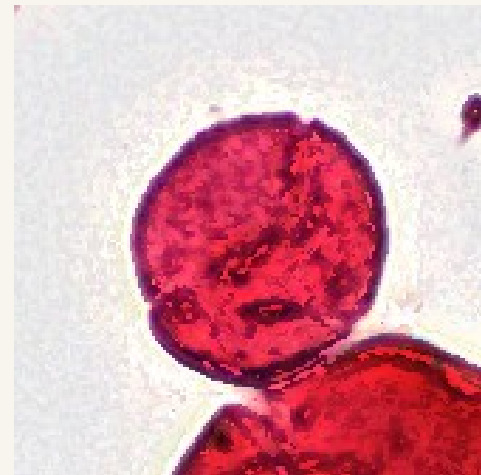


Identifying Fossil Pollen with Machine Learning

Help researchers identify fossil pollen categories more efficiently.



Patrick Batsell · Yun Ying Tsai · Yeonju Kim · Yunfan Bao · Alan Yang

Why fossil pollen matters

01

Why it matters

- Pollen survives in sediment.
- It records ancient plant life.
- We study vegetation about 50 million years ago.

02

The challenge

- Microscope images contain many specimens and debris.
- Identifying each specimen takes expert time.

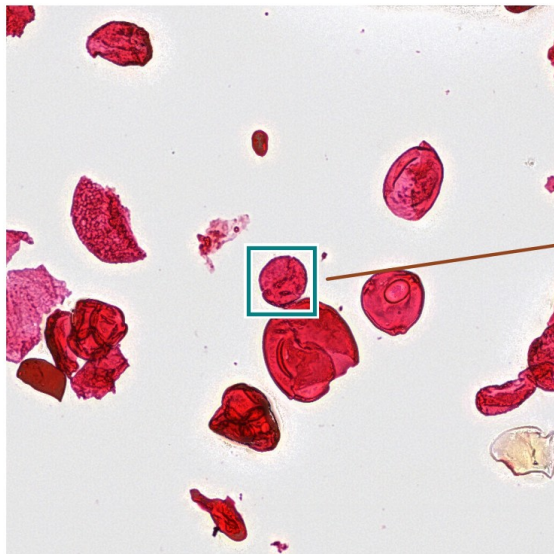
03

Our starting point

- The previous Rice team developed a detector.
- Our next step is identifying the category of each specimen.

One slide, thousands of specimens

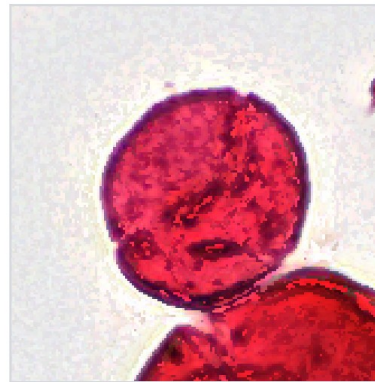
A A crowded microscope image



DETECTION

Where is the specimen?

B Inspect the specimen



Example expert label

Bombacacidites sp.

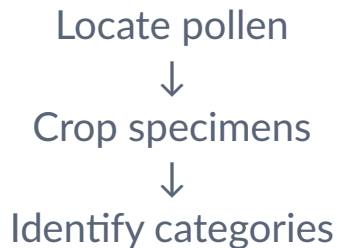
CLASSIFICATION

What type is it?

What earlier studies suggest

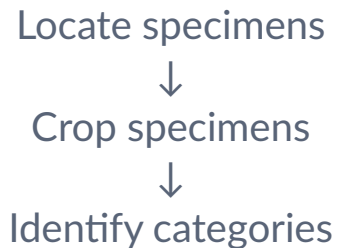
01

Punyasena et al.



02

Martinsen et al.



03

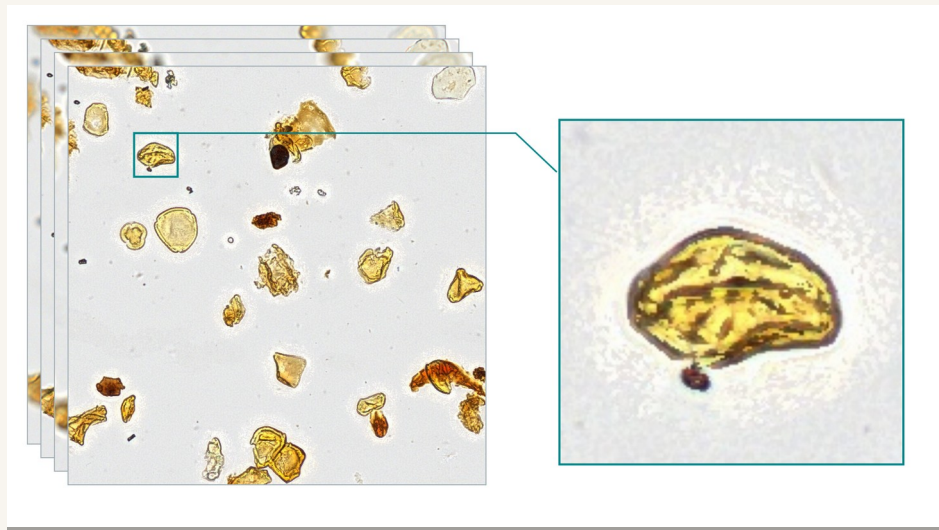
Abbas Shaikh et al.



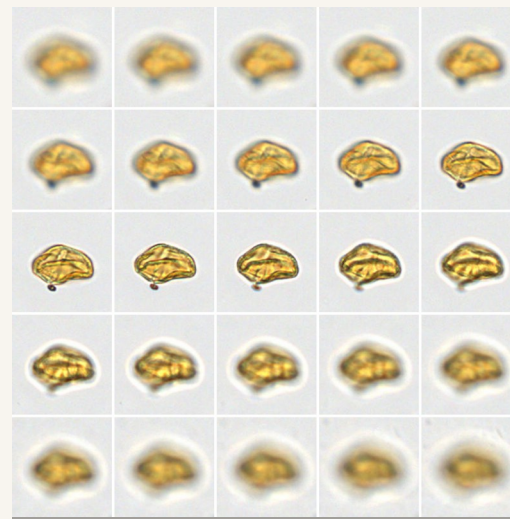
Build on whole-slide detection; add category identification.

One specimen, multiple focal views

- An expert-annotated pollen grain



This example: 25 focal planes
About 30 GB for the full scan



Annotation inventory

83

image–annotation pairs

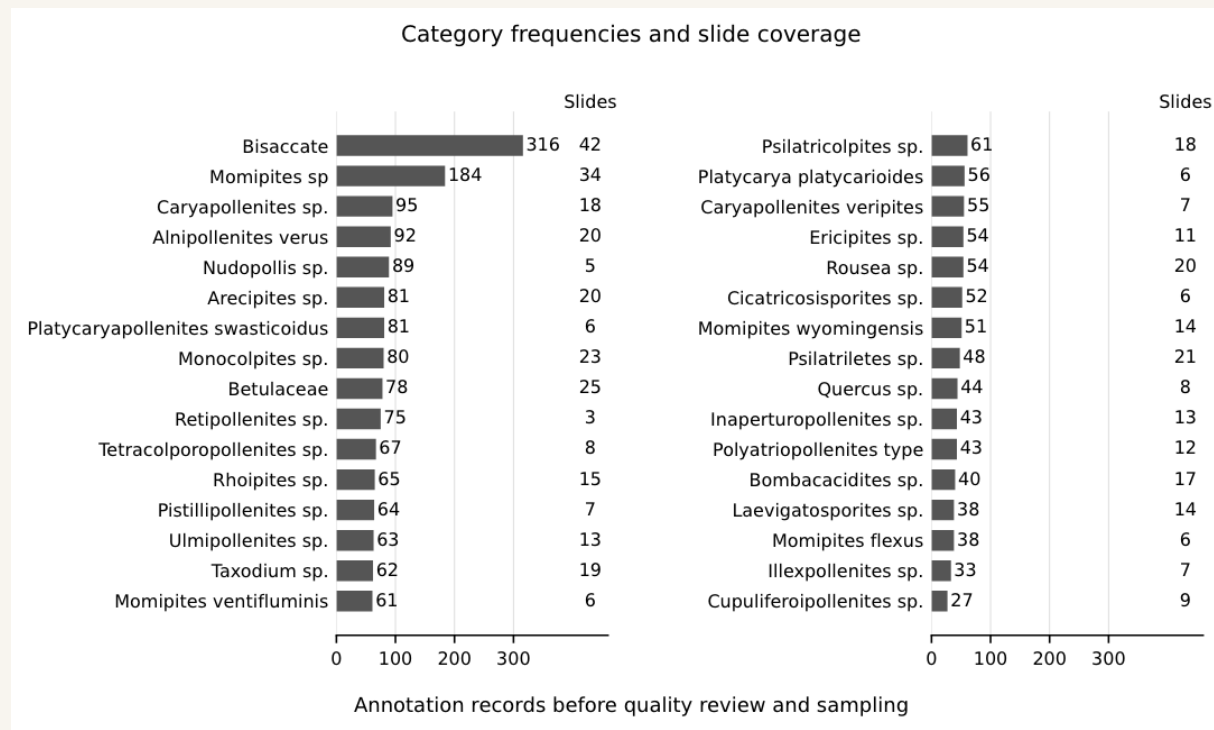
32

priority categories

2,290

annotations

Some categories have
far fewer examples.



Connect images, locations, and expert labels

01

Match the files

Pair each microscope image with its annotation file.



02

Map expert labels

Link original annotation titles to the target categories.



03

Check the examples

Check positions, missing labels, and duplicates.

OUTPUT

Each specimen linked to its image, location, and category

Prepare examples for a fair test

01

Group and split

Keep related slides and every view of a specimen together.



02

Extract and inspect

Extract small image regions; check position and focus.



03

Build model inputs

Regions for detection; specimen crops for classification.

TRAINING ONLY

Vary training examples and balance categories.

Two ways to identify specimens

1 FIND AND IDENTIFY TOGETHER

RF-DETR



Location + category

2 FIND FIRST, IDENTIFY SECOND

RF-DETR

Find specimens



Specimen crops

Isolate each example



Swin-Tiny

Identify its category

Compare location accuracy and category identification.

From finding pollen to identifying it

Compare two approaches to automated identification.

01

Prepare specimen
images

02

Train initial models

03

Evaluate mistakes

Help experts study past vegetation more efficiently.